

HW 3

1. (a) type I error $\rightarrow$ firm is guilty

(reject H_0 | H_0 is true)

$H_0 \rightarrow$ firm is not discriminating

$H_1 \rightarrow$ firm is discriminating

- (b) type II error $\rightarrow$ firm is guilty

(fail to reject H_0 | H_0 is false)

$H_0 \rightarrow$ firm is discriminating

$H_1 \rightarrow$ firm is not discriminating

2. $p = 0.6$: H_0 $n = 10$
 $p < 0.6$: H_1 ≤ 3

- (a) type I $\Rightarrow$ (reject H_0 | H_0 is true)

$$P(p < 0.6 | p = 0.6)$$

$$\alpha = P(X \leq 3 | p = 0.6)$$

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$$

$$X = 0, 1, 2, 3 \quad P(X \leq 3) = 0.0547$$

$$\alpha = 0.0547 \Rightarrow 5.48\%$$

There is 5.48% probability of incorrectly rejecting $p = 0.6$ when the true proportion is 0.6.

- (b) type II $\rightarrow$ (fail to reject H_0 | H_0 is false)

$$P(X \geq 4) \quad \beta(p) = P(X \geq 4 | p)$$

$$= 1 - P(X \leq 3 | p)$$

$$\binom{n}{u} = \frac{n!}{u!(n-u)!}$$

$$p = 0.3 \Rightarrow \beta(0.3) = 1 - \sum_{x=0}^3 \binom{10}{x} (0.3)^x (0.7)^{10-x} = 0.3504$$

$$p = 0.4 \Rightarrow \beta = 0.6177$$

$$p = 0.5 \Rightarrow \beta = 0.8281$$

3. $\mu = 220$, $n = 20$, $\bar{X} = 244$, $\sigma = 24.5$
 $\alpha = 0.05$

$$H_0: \mu = 220$$

$$H_1: \mu > 220 \text{ (right tail)}$$

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

$$t = \frac{244 - 220}{24.5/\sqrt{20}} = \frac{24}{24.5/\sqrt{20}} = \frac{24 \times \sqrt{20}}{24.5} = 4.3808$$

$$t_{0.05, 19} = 1.729$$

$$4.3808 > 1.729 \Rightarrow \text{reject } H_0$$

4. $n_1 = 200$, $\mu_1 = 70750$, $S_1 = 6000$
 $n_2 = 200$, $\mu_2 = 65200$, $S_2 = 5000$
 $\alpha = 0.01$

$$H_0: \mu_1 - \mu_2 = 2000$$

$$H_1: \mu_1 - \mu \neq 2000$$

$$SE = \sqrt{\frac{6000^2}{200} + \frac{5000^2}{200}} = 552.27$$

$$\bar{X}_1 - \bar{X}_2 = 5550$$

$$Z = \frac{5550 - 2000}{552.27}$$

$$Z = 6.43$$

$$Z_{\alpha/2} \approx 2.576$$

$$6.43 > 2.576 \Rightarrow \text{reject } H_0$$

$$0.01$$

5. $X \sim N(1, 0.01^2)$, $\alpha = 0.05$
 $n = 20$, $\bar{X} = 0.992$, $Z_{0.025} = 1.96$

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{992 - 1000}{10/\sqrt{20}} = -3.576$$

$$|-3.578| > 1.96 \Rightarrow \text{fail to reject } H_0$$

$$H_0: \mu = 1000$$

$$H_1: \mu \neq 1000$$

should be stopped

6. C $\rightarrow$ 1000, 1 year
I $\rightarrow$ 500

$\frac{120}{500} \rightarrow$ disease

$\frac{98}{500} \rightarrow$ not disease

$p_1 - p_2 \Rightarrow$ CI

$Z_{0.05} = 1.645$

$$\hat{p} = \frac{120+98}{1000} = \frac{218}{1000}$$

$$\hat{p}_1 = 0.24$$

$$\hat{p}_2 = 0.196$$

$$\hat{p}_1 - \hat{p}_2 = 0.044$$

$$SE = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}} = 0.02608$$

$$\alpha = 0.1, \alpha/2 = 0.05$$

$$Z = 0.044$$

$$Z_{0.05} = 1.645$$

$$ME = 1.645(0.02608) = 0.04290$$

$$CI \Rightarrow (\hat{p}_1 - \hat{p}_2) \pm ME$$

$$\Rightarrow 0.044 \pm 0.04290$$

$$(0.0011, 0.0869)$$

$$p_1 - p_2 > 0 \Rightarrow p_1 > p_2$$

$\therefore$, inoculaⁿ reduces disease incidence

7. $H_0: \sigma^2 = 1$

$n = 5, v = 4$

$H_1: \sigma^2 \neq 1$

$\alpha = 0.05 \Rightarrow \alpha/2 = 0.025$

$1 - \alpha/2 = 0.975$

values $\Rightarrow 1.9, 2.4, 3, 3.5, 4.2$

$$\bar{X} = \frac{\text{sum}}{5} = 3$$

$$S^2 = \frac{\sum (x_i - \bar{X})^2}{n-1} = 0.815$$

$$\chi^2 = \frac{(4)(0.815)}{1}$$

$$\left| \frac{(n-1)S^2}{\chi^2_{\alpha/2, n-1}} < \sigma^2 < \frac{(n-1)S^2}{\chi^2_{1-\alpha/2, n-1}} \right|$$

$(0.29, 7.67)$ | $\notin$ CI
 $\sigma^2 = 1$ is plausible at 95% CI

8. (a) type I $\Rightarrow$ no effect on customer satisfacⁿ

H_0 : effect + or H_0 : no effect
 H_1 : effect no H_1 : +ve effect

type I $\Rightarrow$ {reject H_0 | H_0 is true}
 (there is no effect)

(b) type II $\Rightarrow$ +ve effect

$\Rightarrow$ (fail to reject H_0 | H_0 is false)

$\therefore$, H_0 : +ve effect, H_1 : no effect

9. $H_0: \mu \leq 8$

$n = 225$, $\bar{X} = 8.5$, $SD = 2.25$

$H_1: \mu > 8$

$$SE = \frac{s}{\sqrt{n}} = \frac{2.25}{\sqrt{225}} = \frac{2.25}{15} = 0.15$$

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} = 3.33$$

$$p\text{-value} = P(Z > 3.33) = 0.000429$$

$$p \leq \alpha \Rightarrow \text{reject } H_0 \Rightarrow \mu > 8$$

10. (a) $H_0: \mu = 21.8$, $H_1: \mu \neq 21.8$

(b) $p = 0.2$ $H_0: p \leq 0.2$ $H_1: p > 0.2$

(c) $H_0: \mu = 6.2$, $H_1: \mu \neq 6.2$

(d) $p = 0.7 \Rightarrow H_0: p \geq 0.7$, $H_1: p < 0.7$

(e) $H_0: \mu = 0.58$, $H_1: \mu \neq 0.58$

(f) $H_0: \mu \geq 340$, $H_1: \mu < 340$